

SLR Practice

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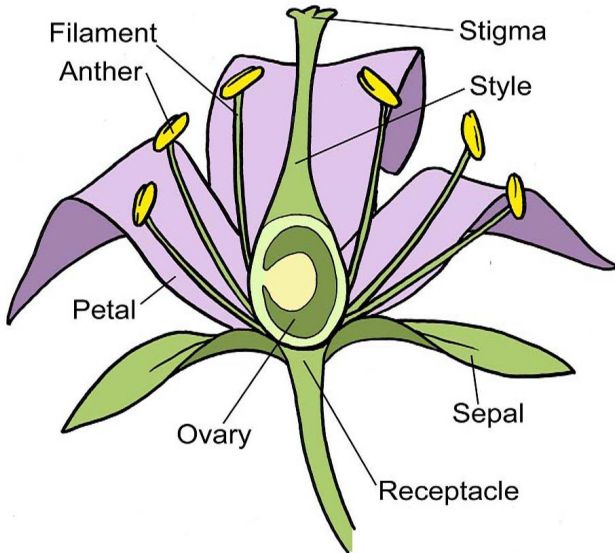


Iris dataset

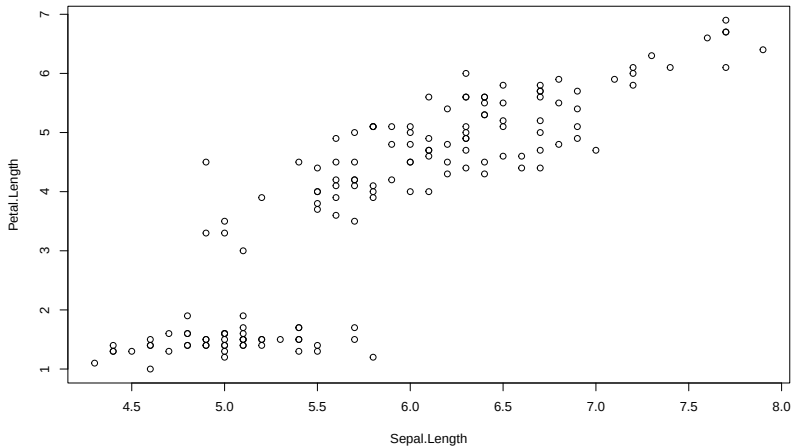
This dataset contains measurements of the Sepals and Petals of 150 iris flowers. The variables in the dataset are:

- Sepal.Length = the length of the sepals in cm.
- Sepal.Width = the width of the sepals in cm.
- Petal.Length = the length of the petals in cm.
- Petal.Width = the width of the petals in cm.
- Species = the species of the iris flower (setosa, versicolor or virginica).

Iris dataset



Iris dataset



Correlation analysis

Calculate the correlation coefficient between sepal length and petal length:

Correlation analysis

Calculate the correlation coefficient between sepal length and petal length:

```
> cor(iris$Sepal.Length,iris$Petal.Length)  
[1] 0.8717538
```

Correlation analysis

Interpret the correlation coefficient:

Correlation analysis

Interpret the correlation coefficient:

There is a strong positive correlation between sepal length and petal length.

Simple Linear Regression Analysis

Use the model output below to answer the following questions:

```
Call:
lm(formula = Petal.Length ~ Sepal.Length, data = iris)

Residuals:
    Min       1Q   Median       3Q      Max
-2.47747 -0.59072 -0.00668  0.60484  2.49512

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -7.10144    0.50666  -14.02  <2e-16 ***
Sepal.Length   1.85843    0.08586   21.65  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8678 on 148 degrees of freedom
Multiple R-squared:  0.76,    Adjusted R-squared:  0.7583
F-statistic: 468.6 on 1 and 148 DF,  p-value: < 2.2e-16
```

1. Give the estimated regression equation from the model output (include the actual estimated values of the coefficients).

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1. Give the estimated regression equation from the model output (include the actual estimated values of the coefficients).

$$\widehat{Petal.Length}_i = -7.10 + 1.86(Sepal.Length)$$

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2. Interpret the Sepal.Length coefficient.

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2. Interpret the Sepal.Length coefficient.

On average, the petal length is 1.86 cm longer for each 1cm increase in sepal length.

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3. Conduct a hypothesis test for the Sepal.Length coefficient.

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3. Conduct a hypothesis test for the Sepal.Length coefficient.

$$H_0: \beta_1 = 0$$

$$H_1: \beta_1 \neq 0$$

$$\alpha = 0.05$$

$$t_{\text{stat}} = 21.65$$

$$p\text{-value} < 0.001$$

conclusion: reject H_0 because p-value is less than 0.05. There is a linear relationship between iris petal length and sepal length.

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4. What proportion of variance in petal length is explained by sepal length?

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5. What does the residual standard error of the model measure?

Simple Linear Regression Analysis

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```

5. What does the residual standard error of the model measure?

The standard deviation of the model residuals.

Simple Linear Regression Analysis

```
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6. What is the predicted petal length when sepal length is 5.3cm?

Simple Linear Regression Analysis

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6. What is the predicted petal length when sepal length is 5.3cm?

$$\widehat{Petal.Length}_i = -7.10144 + 1.85843(Sepal.Length)$$

$$\widehat{Petal.Length}_i = -7.10144 + 1.85843(5.3)$$

$$\widehat{Petal.Length}_i = 2.75$$

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7. What is the confidence interval for the predicted petal length when sepal length is 5.3cm? The mean of sepal length is 5.843333 and the sum of squares for sepal length is 102.1683.

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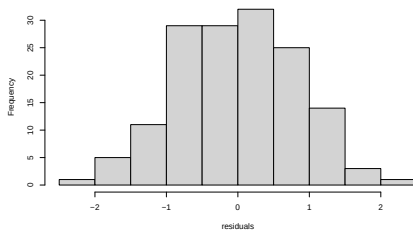
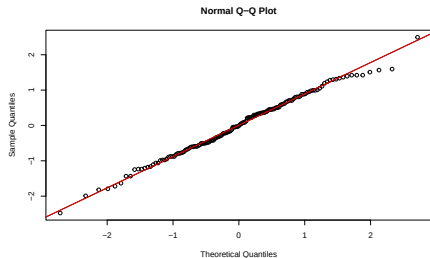
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$$\hat{y} \pm t_{\frac{\alpha}{2}, n-2} s_e \sqrt{\frac{1}{n} + \frac{(x_p - \bar{x})^2}{\sum_i (x_i - \bar{x})^2}}$$

$$2.748239 \pm 1.976 \times 0.8678 \sqrt{\frac{1}{150} + \frac{(5.3 - 5.843333)^2}{102.1683}}$$

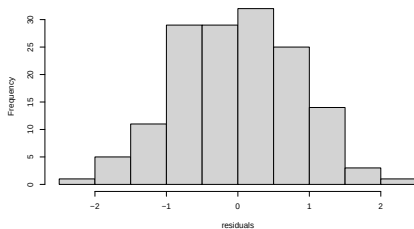
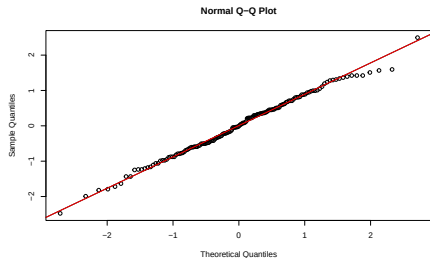
$$[2.58, 2.91]$$

Simple Linear Regression Analysis



8. What model assumption are the above plots used to test and explain whether this assumption is met?

Simple Linear Regression Analysis



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These plots are used to test the assumption that the model residuals are normally distributed. This assumption is met as the distribution shown in the histogram appears normal and the points follow the line on the qqplot well, except for a few expected deviations in the tails.